

Delivery Plan — Kesho Chama App

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Generated: 14 August 2026

Planned start: 01 September 2026

Projected delivery: 03 December 2026

Duration: 67 working days (~13.4 weeks)

Total effort: 84.0 person-days across 30 tasks

Team: 8 people — 2× Backend Engineer, 1× DevOps Engineer, 2× Frontend Engineer, 1× Product Manager, 1× QA Engineer, 1× UX Designer

Dates are computed from the task graph, not generated by the language model: dependencies and per-role capacity are resolved in code, so the timeline is arithmetically consistent with the estimates above it.

Executive Summary

Kesho is building a digital savings app for informal-sector workers in Kenya, allowing groups to manage their chamas digitally. The app will track contributions, predict defaults, and provide a reputation system. The target users are market traders, boda riders, and others who earn daily in cash and have limited banking relationships.

Business objectives

- Provide a reliable and secure digital platform for informal savings groups
- Reduce disputes and increase trust among group members
- Comply with relevant regulatory requirements

Scope

In scope

- Development of a mobile app for Android devices
- Integration with M-Pesa for money transactions
- Contribution tracking and payout management
- Group admin features for monitoring and chasing late payments
- Reputation system for members
- Support for English and Swahili languages

Explicitly out of scope

- Banking or financial services beyond digital chama management
- Hardware or infrastructure provision for users
- Comprehensive user training or support programs

Open Questions

These need client answers. Each shows the assumption the plan currently runs on, so the cost of a different answer is visible.

#	Question	Why it matters	Assumed for now
1	What is the expected average group size and number of groups to be supported by the app?	Scalability and performance requirements will depend on the number of users and groups	Assuming an average group size of 10-20 members and 100-500 groups initially
2	How will the app handle cases where a member disputes a transaction or contribution record?	Dispute resolution processes will impact the app's functionality and user trust	Assuming a basic dispute resolution process will be implemented, with potential for escalation to group admins or external authorities

#	Question	Why it matters	Assumed for now
3	What specific regulatory requirements or guidelines from the Central Bank of Kenya (CBK) need to be complied with?	Non-compliance with regulatory requirements could result in legal or financial penalties	Assuming compliance with general guidelines for non-banking financial services, but requiring further clarification from the client
4	How will the AI-powered default prediction feature be trained and validated, and what data will be used for this purpose?	The accuracy and effectiveness of the default prediction feature will depend on the quality and relevance of the training data	Assuming that historical contribution data will be used for training, but requiring further discussion with the client on data availability and quality
5	What is the expected timeline for the seed round, and what are the key milestones or deliverables required before then?	The project timeline and milestones will impact the development schedule and resource allocation	Assuming a four-month timeline with key milestones including alpha and beta releases, but requiring further clarification from the client

Risks

Severity	Risk	Mitigation
HIGH	Non-compliance with regulatory requirements	Close collaboration with the client to understand regulatory requirements, and engagement with relevant experts or authorities as needed
MEDIUM	Technical difficulties with M-Pesa integration	Early engagement with M-Pesa technical teams, and thorough testing of integration components

Task Breakdown

E1 — Discovery and Planning

Define project scope, timeline, and technical requirements

3 tasks · 7 person-days

ID	Task	Role	Est.	Depends on	Start	End	Critical
T1	Conduct stakeholder interviews	Product Manager	2.0d	—	01 Sep	03 Sep	yes
T2	Define project scope and timeline	Product Manager	3.0d	T1	03 Sep	08 Sep	yes
T22	Define dispute resolution process	Product Manager	2.0d	T1	08 Sep	10 Sep	yes

E2 — Design

Create user interface and user experience designs for the mobile app

2 tasks · 8 person-days

ID	Task	Role	Est.	Depends on	Start	End	Critical
T3	Create wireframes and prototypes	UX Designer	5.0d	T2	08 Sep	15 Sep	yes
T4	Design the visual identity	UX Designer	3.0d	T3	15 Sep	18 Sep	yes

E3 — Mobile App Development

Build the mobile app for Android devices

6 tasks · 21 person-days

ID	Task	Role	Est.	Depends on	Start	End	Critical
T5	Build the mobile app framework	Mobile Engineer	5.0d	T4	18 Sep	25 Sep	yes
T6	Implement contribution tracking	Mobile Engineer	4.0d	T5	25 Sep	01 Oct	yes
T7	Implement group admin features	Mobile Engineer	4.0d	T6	01 Oct	07 Oct	yes
T8	Implement reputation system	Mobile Engineer	3.0d	T7	07 Oct	12 Oct	yes
T20	Implement language support	Mobile Engineer	2.0d	T8	12 Oct	14 Oct	yes
T23	Implement dispute resolution feature	Mobile Engineer	3.0d	T22	10 Sep	15 Sep	yes

E4 — Integration with M-Pesa

Integrate the app with M-Pesa for money transactions

2 tasks · 9 person-days

ID	Task	Role	Est.	Depends on	Start	End	Critical
T9	Integrate with M-Pesa API	Backend Engineer	5.0d	T8	12 Oct	19 Oct	yes
T10	Implement transaction handling	Backend Engineer	4.0d	T9	19 Oct	23 Oct	yes

E5 — Testing and Quality Assurance

Ensure the app meets the required quality and functionality standards

7 tasks · 17 person-days

ID	Task	Role	Est.	Depends on	Start	End	Critical
T11	Develop unit tests	Mobile Engineer	3.0d	T10	23 Oct	28 Oct	yes
T12	Develop integration tests	Mobile Engineer	3.0d	T11	28 Oct	02 Nov	yes
T13	Conduct manual testing	QA Engineer	4.0d	T12	02 Nov	06 Nov	yes
T21	Test language support	QA Engineer	1.0d	T20	14 Oct	15 Oct	
T24	Test dispute resolution feature	QA Engineer	1.0d	T23	15 Sep	16 Sep	
T29	Conduct alpha testing	QA Engineer	2.0d	T13	06 Nov	10 Nov	yes
T30	Conduct beta testing	QA Engineer	3.0d	T29	10 Nov	13 Nov	

E6 — Security and Compliance

Ensure the app meets security and compliance requirements

5 tasks · 14 person-days

ID	Task	Role	Est.	Depends on	Start	End	Critical
T14	Conduct security audit	Security Engineer	3.0d	T13	06 Nov	11 Nov	yes
T15	Implement security measures	Security Engineer	4.0d	T14	11 Nov	17 Nov	yes
T16	Ensure compliance with regulations	Security Engineer	2.0d	T15	19 Nov	23 Nov	yes
T25	Conduct data protection impact assessment	Security Engineer	2.0d	T14	17 Nov	19 Nov	yes
T26	Implement data protection measures	Security Engineer	3.0d	T25	23 Nov	26 Nov	

E7 — Deployment and Launch

Deploy the app and launch it to the public

5 tasks · 8 person-days

ID	Task	Role	Est.	Depends on	Start	End	Critical
T17	Prepare for deployment	DevOps Engineer	2.0d	T16	23 Nov	25 Nov	yes
T18	Deploy the app	DevOps Engineer	1.0d	T17	25 Nov	26 Nov	yes
T19	Launch the app	Product Manager	2.0d	T18	26 Nov	30 Nov	yes
T27	Prepare marketing materials	Product Manager	2.0d	T19	30 Nov	02 Dec	yes
T28	Coordinate app launch	Product Manager	1.0d	T27	02 Dec	03 Dec	yes

Timeline

Critical path: **26 of 30 tasks** have zero slack. Any slip on those moves the delivery date directly.

Task	Role	Start	End	Slack
T1 Conduct stakeholder interviews	Product Manager	01 Sep 2026	03 Sep 2026	0
T2 Define project scope and timeline	Product Manager	03 Sep 2026	08 Sep 2026	0
T22 Define dispute resolution process	Product Manager	08 Sep 2026	10 Sep 2026	0
T3 Create wireframes and prototypes	UX Designer	08 Sep 2026	15 Sep 2026	0
T23 Implement dispute resolution feature	Mobile Engineer	10 Sep 2026	15 Sep 2026	0

Task	Role	Start	End	Slack
T4 Design the visual identity	UX Designer	15 Sep 2026	18 Sep 2026	0
T5 Build the mobile app framework	Mobile Engineer	18 Sep 2026	25 Sep 2026	0
T6 Implement contribution tracking	Mobile Engineer	25 Sep 2026	01 Oct 2026	0
T7 Implement group admin features	Mobile Engineer	01 Oct 2026	07 Oct 2026	0
T8 Implement reputation system	Mobile Engineer	07 Oct 2026	12 Oct 2026	0
T20 Implement language support	Mobile Engineer	12 Oct 2026	14 Oct 2026	0
T9 Integrate with M-Pesa API	Backend Engineer	12 Oct 2026	19 Oct 2026	0
T10 Implement transaction handling	Backend Engineer	19 Oct 2026	23 Oct 2026	0
T11 Develop unit tests	Mobile Engineer	23 Oct 2026	28 Oct 2026	0
T12 Develop integration tests	Mobile Engineer	28 Oct 2026	02 Nov 2026	0
T13 Conduct manual testing	QA Engineer	02 Nov 2026	06 Nov 2026	0
T14 Conduct security audit	Security Engineer	06 Nov 2026	11 Nov 2026	0
T29 Conduct alpha testing	QA Engineer	06 Nov 2026	10 Nov 2026	0
T15 Implement security measures	Security Engineer	11 Nov 2026	17 Nov 2026	0
T25 Conduct data protection impact assessment	Security Engineer	17 Nov 2026	19 Nov 2026	0
T16 Ensure compliance with regulations	Security Engineer	19 Nov 2026	23 Nov 2026	0
T17 Prepare for deployment	DevOps Engineer	23 Nov 2026	25 Nov 2026	0
T18 Deploy the app	DevOps Engineer	25 Nov 2026	26 Nov 2026	0
T19 Launch the app	Product Manager	26 Nov 2026	30 Nov 2026	0
T27 Prepare marketing materials	Product Manager	30 Nov 2026	02 Dec 2026	0
T28 Coordinate app launch	Product Manager	02 Dec 2026	03 Dec 2026	0

Effort by role — total 84.0 person-days, compressed into 67 working days (1.25× average parallelism).

Role	Person-days
Backend Engineer	9.0
DevOps Engineer	3.0
Mobile Engineer	27.0
Product Manager	12.0
QA Engineer	11.0
Security Engineer	14.0
UX Designer	8.0

Recommended Tech Stack

Layer	Choice	Why	Runner-up	Why not
Frontend	React Native	Cross-platform compatibility, large community support, and ease of development for Android devices, which is the primary target for this project	Flutter	While Flutter is a strong alternative, React Native's larger community and more extensive library support make it a better fit for this project's medium complexity and the need for rapid development
Backend	Node.js with Express.js	Well-established framework for building scalable and secure server-side applications, with extensive support for integrating with third-party services like M-Pesa	Django	Django, while robust, has a steeper learning curve and is more opinionated, which could slow down development for a small team
Database	PostgreSQL	Reliable, scalable, and feature-rich open-source database that supports complex queries and has strong support for transactions, necessary for managing financial data	MySQL	PostgreSQL offers better support for advanced data types and querying capabilities, making it more suitable for the financial tracking and prediction aspects of the app
Hosting	AWS	Comprehensive set of services that can support the application's needs, from compute and storage to database and security, with a strong presence in Africa for low latency	Google Cloud Platform (GCP)	AWS has a more established presence in Africa and offers more competitive pricing for small to medium-sized projects, making it a better fit for this startup
Auth	OAuth 2.0 with Okta	Secure, scalable, and widely adopted authorization framework that can handle the app's authentication needs, including support for M-Pesa integration	Auth0	Okta offers more comprehensive features for identity and access management, particularly for enterprise and financial applications, making it a better choice for compliance and security
CI/CD	Jenkins	Mature, open-source automation server that supports a wide range of development lifecycles, including build, test, and deployment, with a large community for support	GitLab CI/CD	Jenkins offers more flexibility and customization options, which are beneficial for integrating with the diverse set of tools and services used in this project
Monitoring	Prometheus and Grafana	Prometheus provides robust monitoring and alerting capabilities, while Grafana offers flexible and customizable dashboards for visualizing application performance and health	New Relic	The combination of Prometheus and Grafana offers more cost-effective and customizable monitoring solutions, which are crucial for a startup's budget and specific needs
M-Pesa Integration	Daraja API by Safaricom	Official API provided by Safaricom for integrating M-Pesa services into applications, ensuring compliance and reliability for financial transactions	Third-party M-Pesa SDKs	Using the official Daraja API ensures direct support from Safaricom and compliance with regulatory requirements, reducing the risk of integration issues or service disruptions

Architecture

- The React Native frontend will communicate with the Node.js backend via RESTful APIs, secured with OAuth 2.0 and Okta for authentication.
- The backend will utilize PostgreSQL for database operations, ensuring data integrity and scalability.
- M-Pesa integration will be handled through the Daraja API, facilitating secure and compliant financial transactions.
- Monitoring will be achieved through Prometheus and Grafana, providing real-time insights into application performance and health.

Scaling considerations

- As usage grows, the database (PostgreSQL) will likely be the first component to require scaling, which can be addressed by upgrading to more powerful instances or implementing sharding.
- The Node.js backend may also require additional instances or more efficient routing to handle increased traffic, which can be managed through AWS auto-scaling features.
- To mitigate the risk of non-compliance with regulatory requirements, regular audits and compliance checks will be necessary, potentially involving third-party services for support.

Plan Validation

The model's task graph passed validation with no repairs: all task ids unique, all dependencies resolvable, all estimates within range.